education

Department:
Education
REPUBLIC OF SOUTH AFRICA

NATURAL SCIENCE MARKING GUIDELINES

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MATHEMATICS N5
16030175

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WISKUNDE N5

Question / Vraag 1

2010-01-24

1.1.1 $\lim\limits_{x\to\frac{\pi}{4}} \left(\frac{1-\tan^2 x}{\frac{\pi}{8}-\frac{1}{2}x}\right) = \frac{0}{0}$ .... indeterminate / onbepaald

$= \lim\limits_{x\to\frac{\pi}{4}} \frac{-2\tan x.\sec^2 x}{-\frac{1}{2}}$ ✓ ✓

$= \frac{-2\tan\left(\frac{\pi}{4}\right).\frac{1}{(\cos^2\frac{\pi}{4})}}{-\frac{1}{2}}$ ✓ (4)

$= 8$ ✓

1.1.2 $\lim\limits_{x\to 2} \left(\frac{x-2}{\ln\left(\frac{x}{2}\right)}\right) = \frac{0}{0}$ .... indeterminate / onbepaald

$= \lim\limits_{x\to 2} \frac{1}{\frac{2}{x}.\frac{1}{2}}$ ✓ ✓

$= \lim\limits_{x\to 2} x$ (2)

$= 2$ ✓

1.2 For discontinuity / Vir diskontinuïteit :

$x^2+5x+6=0$

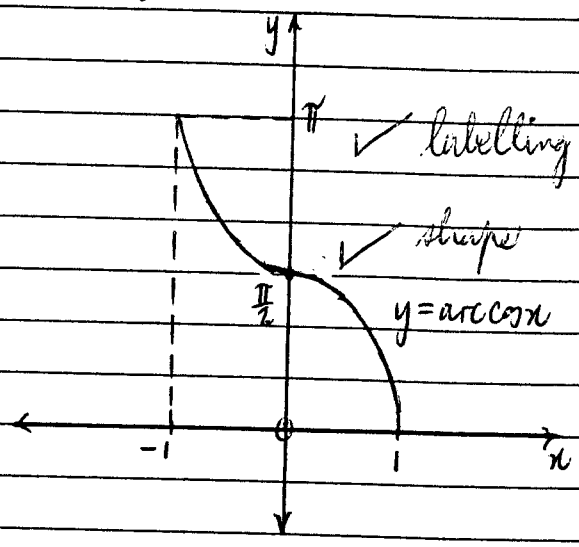
$(x+3)(x+2)=0$ (2)

$\therefore x=-3$ ✓ and / en $x=-2$ ✓

[8]

Question / Vraag 2

2.1.1



✓ labelling
✓ shape
(2)

2.1.2 $y = \arccos x$

$\cos y = x$

$-\sin y \cdot \frac{dy}{dx} = 1$ ✓

$\frac{dy}{dx} = -\frac{1}{\sin y}$ ✓ where / waar $\sin y = \sqrt{1-\cos^2 y}$ (3)

$= -\frac{1}{\sqrt{1-\cos^2 y}}$ ✓ where / waar $\cos y = x$

$\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$

2.2 $y = \tan x = \frac{\sin x}{\cos x}$

$\frac{dy}{dx} = \frac{(\cos x)(\cos x) - (-\sin x)(\sin x)}{(\cos x)^2}$ ✓

$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$ ✓ (3)

$\frac{dy}{dx} = \frac{1}{\cos^2 x}$ ✓ $= \sec^2 x$

2.3.1 $\frac{dy}{dx} = \frac{(e^x)(\ln^2 x) - (2\ln x \cdot \frac{1}{x})(e^x)}{(\ln^2 x)^2}$ ✓ ✓ ✓ (3)

2.3.2 $\frac{dy}{dx} = (3x^2)(\text{arc cot } x) + \left(\frac{-1}{1+x^2}\right)(x^3)$ ✓ P.R. ✓ ✓ (3)

2.3.3 $\frac{dy}{dx} = \frac{1}{4}(x^4 + \sec 7x)^{-\frac{3}{4}} \cdot (4x^3 + \sec 7x \cdot \tan 7x \cdot 7)$ ✓ ✓ ✓ (3)

2.4 $\frac{1}{y^2} - \frac{1}{x^2} = y + x$

$-2y^{-3} \cdot \frac{dy}{dx} + 2x^{-3} = 1 \cdot \frac{dy}{dx} + 1$ ✓ ✓

$\frac{dy}{dx}(-2y^{-3} - 1) = 1 - 2x^{-3}$ ✓ (4)

$\frac{dy}{dx} = \frac{1 - 2x^{-3}}{-2y^{-3} - 1} = \frac{1 - 2(1)^{-3}}{-2(2)^{-3} - 1} = \frac{4}{5}$ ✓

2.5 $y = 4^x \cdot e^{-x}$

$\ln y = \ln(4^x \cdot e^{-x})$

$= \ln 4^x + \ln e^{-x}$

$\ln y = x\ln 4 - x$

$\frac{1}{y} \cdot \frac{dy}{dx} = 1 \cdot \ln 4 - 1$ ✓ ✓ ✓ (4)

$\frac{dy}{dx} = y(\ln 4 - 1)$

$\frac{dy}{dx} = 4^x \cdot e^{-x}(\ln 4 - 1)$

[25]

Question/Vraag 3

3.1 $h_s + h_1 = 20$ $\quad h_s^2 = h_1^2 + r^2$

$\therefore h_s = 20 - h_1$ ✓ $\quad (20 - h_1)^2 = h_1^2 + r^2$

$400 - 40h_1 + h_1^2 = h_1^2 + r^2$

$\therefore r^2 = 400 - 40h_1$ ✓

$V = \frac{1}{3}\pi r^2 h_1$

$= \frac{1}{3}\pi (400 - 40h_1)h_1$

$V = \frac{1}{3}\pi (400h_1 - 40h_1^2)$ ✓ (6)

$\frac{dV}{dh_1} = \frac{1}{3}\pi (400 - 80h_1) = 0$ ✓

$\therefore h_1 = 5$ cm ✓

$h_s = 20 - 5 = 15$ cm ✓

$r = \sqrt{400 - 40(5)} = 14,142$ cm ✓

3.2.1 $S = 3e^{-2,5t} \cdot \cos 2\pi t$ after/na $t = 2$s

$S = 3e^{-2,5(2)} \cdot \cos 2\pi . 2$ ✓

$S = 0,020$ cm ✓ (2)

3.2.2 $V = \frac{ds}{dt} = (-7,5e^{-2,5t})(\cos 2\pi t) + (-2\pi \sin 2\pi t)(3e^{-2,5t})$ ✓ ✓ (2)

3.3 $f(x) = -x^3 - 2x^2 + 3x + 1$

3.3.1 $f'(x) = -3x^2 - 4x + 3 = 0$

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(-3)(3)}}{2(-3)} = \frac{4 \pm \sqrt{52}}{-6}$$

$\therefore x = -1,869$ and/en $x = 0,535$

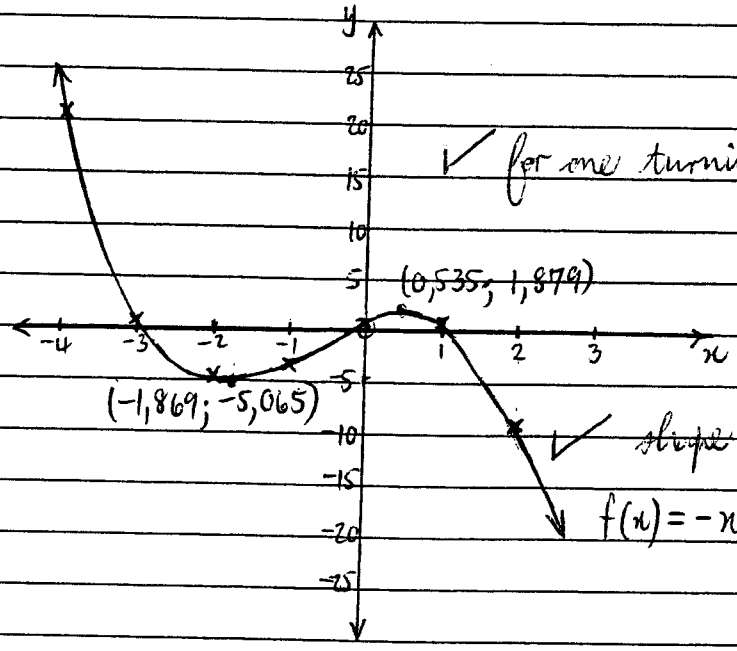
$\therefore$ if/as $x = -1,869$; $y = -5,065$ ✓
if/as $x = 0,535$; $y = 1,879$ ✓ (2)

3.3.2

x	-4	-3	-2	-1	0	1	2
$f(x)$	21	1	-5	-3	1	1	-9

✓✓ (2)

3.3.3



✓ for one turning point
✓ shape
(2)

3.3.4 let/laat $x_1 = -0,3$ $\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$f(-0,3) = -0,053$ ✓
$f'(-0,3) = 3,930$ ✓

$= -0,3 - \left(\frac{-0,053}{3,930}\right)$ ✓

$x_2 = -0,287$ ✓ (4)

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Question / Vraag 4

4.1.1 $\int e^{2x}.\sin x.dx = e^{2x}.-\cos x - \int 2e^{2x}.-\cos x.dx$ ✓

$= -e^{2x}.\cos x + 2\int e^{2x}.\cos x.dx$

$= -e^{2x}.\cos x + 2\left(e^{2x}.\sin x - \int 2e^{2x}.\sin x.dx\right)$ ✓

$\int e^{2x}.\sin x.dx = -e^{2x}\cos x + 2e^{2x}\sin x - 4\int e^{2x}\sin x\,dx$

$5\int e^{2x}.\sin x.dx = -e^{2x}\cos x + 2e^{2x}\sin x$ ✓ (4)

$\therefore \int e^{2x}\sin x.dx = -\frac{1}{5}e^{2x}\cos x + \frac{2}{5}e^{2x}\sin x + C$ ✓ ✓

4.1.2 $\int \frac{\pi^{\ln x}}{x}.dx$

$= \int \pi^{\ln x}.\frac{1}{x}dx$

$= \int \pi^u.du$ ✓

$= \frac{\pi^u}{\ln\pi} + C = \frac{\pi^{\ln x}}{\ln\pi} + C$ ✓ (3)

$u = \ln x$
$du = \frac{1}{x}.dx$ ✓

4.1.3 $\int \frac{1}{25+64x^2}\,.dx$

$= \int \frac{1}{64\left(\frac{25}{64}+x^2\right)}\,.dx$ ✓

$= \frac{1}{64}\int \frac{1}{\frac{25}{64}+x^2}\,.dx$ (3)

$= \frac{1}{64}\,.\,\frac{1}{\frac{5}{8}}\arctan\left(\frac{x}{\frac{5}{8}}\right)+C$ ✓ ✓

$= \frac{1}{40}\arctan\left(\frac{8x}{5}\right)+C$ →

4.1.4 $\int \cos 7x.\sin 5x.dx$

$= \frac{1}{2}\int (\sin 12x - \sin 2x).dx$ ✓

$= \frac{1}{2}\left(-\frac{\cos 12x}{12}+\frac{\cos 2x}{2}\right)+C$ (3)

$= -\frac{\cos 12x}{24}+\frac{\cos 2x}{4}+C$ →

4.1.5 $\int \frac{\cos 2x}{\sin x+\cos x}\,.dx$

$= \int \frac{\cos^2 x-\sin^2 x}{\sin x+\cos x}\,.dx$ ✓

$= \int \frac{(\cos x+\sin x)(\cos x-\sin x)}{(\sin x+\cos x)}\,.dx$ ✓ (4)

$= \int (\cos x-\sin x).dx$

$= \sin x+\cos x+C$ ✓ ✓ →

4.2 $\dfrac{x+1}{x^2-10x+25} = \dfrac{x+1}{(x-5)^2} = \dfrac{A}{(x-5)^2} + \dfrac{B}{(x-5)}$

$\therefore \quad x+1 = A(1) + B(x-5)$ ✓

$x=5: \quad 6 = A$ ✓

(5)

coeff. of x: $1 = B$ ✓

$$\int \frac{x+1}{x^2-10x+25}\,dx = 6\int \frac{1}{(x-5)^2}\,dx + \int \frac{1}{(x-5)}\,dx$$

$$= -\frac{6}{(x-5)} + \ln(x-5) + C$$ ✓ ✓

[22]

Question / Vraag 5

5.1 $$\int_0^{\infty} e^{-st} \cdot -\frac{1}{4}\,dt$$

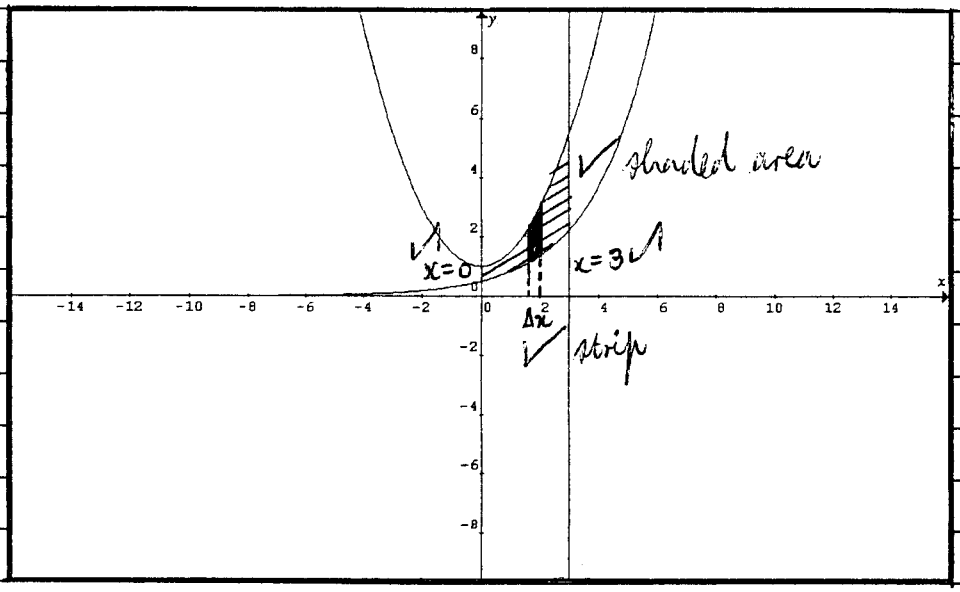
$$= \frac{1}{4s}\left(e^{-st}\right)\Big|_0^{\infty}$$ ✓

$$= \frac{1}{4s}\left(e^{-\infty} - e^{0}\right)$$ ✓ (4)

$$= \frac{1}{4s}\left(0 - 1\right)$$ ✓

$$= -\frac{1}{4s}$$ ✓

5.2.1



(3)

5.2.2 $\delta A = (y_2 - y_1)\,\delta x$

$\delta A = \left(\left(\frac{1}{2}x^2+1\right) - \left(\frac{1}{2}e^{\frac{1}{2}x}\right)\right)\delta x$

$\therefore A = \int_0^3 \left(\frac{1}{2}x^2+1-\frac{1}{2}e^{\frac{1}{2}x}\right).dx$

$= \left(\frac{1}{6}x^3 + x - e^{\frac{1}{2}x}\right)\Big|_0^3$ (3)

$A = 4,018 \text{ units}^2$

5.2.3 $\delta V = \pi(y_2^2 - y_1^2)\,\delta x$

$\delta V = \pi\left[\left(\frac{1}{2}x^2+1\right)^2 - \left(\frac{1}{2}e^{\frac{1}{2}x}\right)^2\right]\delta x$

$\therefore V = \pi\int_0^3 \left(\frac{1}{4}x^4 + x^2 + 1 - \frac{1}{4}e^x\right).dx$ (4)

$= \pi\left(\frac{1}{20}x^5 + \frac{1}{3}x^3 + x - \frac{1}{4}e^x\right)\Big|_0^3$

$V = 60,880 \text{ units}^3$

5.3 Mass per unit area / Massa per area eenheid $= \frac{m}{ab}$

$\therefore \Delta m = b \times \Delta x \times \frac{m}{ab} = \frac{m}{a} . \Delta x$ ✓

$\therefore \Delta I = \Delta m \times x^2$

$= \frac{m}{a} . \Delta x \times x^2$ ✓

$\therefore I = \frac{m}{a} \int_0^a x^2 . dx$ (4)

$= \frac{m}{a} \left[\frac{x^3}{3} \right]_0^a$ ✓

$\underline{I = \frac{1}{3} m a^2 \text{ kg. units}^2}$ ✓ →

[18]

Question / Vraag 6

6.1 $\frac{dy}{dx} = \frac{3x+1}{3y}$

$\int 3y . dy = \int (3x+1) . dx$ ✓ ✓

$\frac{3}{2} y^2 = \frac{3}{2} x^2 + x + C$ ✓ ✓ (4)

$\frac{3}{2} (1)^2 = \frac{3}{2} (-1)^2 + (-1) + C$

$\therefore C = 1$ ✓

$\underline{\frac{3}{2} y^2 = \frac{3}{2} x^2 + x + 1}$ →

6.2 $\frac{d^2y}{dx^2} = (e^{-x}+1)^2$

$\frac{d^2y}{dx^2} = e^{-2x} + 2e^{-x} + 1$

$\frac{dy}{dx} = -\frac{1}{2}e^{-2x} - 2e^{-x} + x + C_1$ (3)

$y = \frac{1}{4}e^{-2x} + 2e^{-x} + \frac{1}{2}x^2 + C_1x + C_2$

[7]

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